McMullen Maps

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The aim of this document is to prove that the reduction techniques of Faber and Voloch used to study Newton maps of polynomials can also be used to study McMullen maps of cubics. Concretely, outside of a finite set of primes p to be determined, the reduced map modulo p reaching a root of (f mod p) is sufficient to conclude p-adic convergence of the iterates (x_n) of T_f. However, unlike with Newton maps, this is not a necessary condition for convergence.

1 Definitions

Firstly, we shall make some definitions. Let $f(x) = x^3 + ax + b$. We will assume that[1] $\Delta(f) = -4a^3 - 27b^2 \neq 0$. Then the McMullen Map of f is defined to be

$$T_f(x) = x - \frac{(x^3 + ax + b)(3ax^2 + 9bx - a^2)}{3ax^4 + 18bx^3 - 6a^2x^2 - 6abx - 9b^2 - a^3}. \tag{1}$$

This is the Newton map of the rational function

$$q(x) = \frac{f(x)}{h(x)},$$

where $h(x) = 3ax^2 + 9bx - a^2$. This algorithm succeeds because the points of inflection of q are precisely the roots of f, so by Thereom 1 of Smale (1985), we have convergence on an open dense set of initial points.

Let α be a root of f, and define $g_\alpha(x) = x^2 + \alpha x + a + \alpha^2$, so $f(x) = (x - \alpha)g_\alpha(x)$. The following computation will simplify our algebra later.

$$\begin{aligned} h(x)g'_\alpha(x) - g_\alpha(x)h'(x) &= (3ax^2 + 9bx - a^2)(2x + \alpha) - (x^2 + \alpha x + a + \alpha^2)(6ax + 9b) \\ &= 3(3b - a\alpha)x^2 - 2a(4a + 3\alpha^2)x - (9b\alpha^2 + a^2\alpha + 9ab) \\ &= (x - \alpha)\underbrace{\left(3(3b - a\alpha)x - (9a\alpha^2 - 9b\alpha + 8a^2)\right)}_{k_\alpha(x)}. \end{aligned}$$

[1] if $\Delta(f) = 0$ then q is linear and thus T_f is constant.

2 Recreation of Faber-Voloch Proofs

Proposition 1. *Let $f(x) = x^3 + ax + b \in L[x]$ with a, b not both zero. If $\Delta(f) = -4a^3 - 27b^2 = 0$, then T_f is a constant map, where the constant is the simple root of f. Otherwise, T_f is a rational map of degree 4.*

Proof. For the first part of the proof, suppose $\Delta(f) = 0$. Then $f(x)$ takes the form

$$f(x) = (x - \alpha)^2(x + 2\alpha) = x^3 - 3\alpha^2 x + 2\alpha^3.$$

It follows that

$$h(x) = -9\alpha^2(x - \alpha)^2, \quad q(x) = -\frac{x + 2\alpha}{9\alpha^2}, \quad T_f(x) = -2\alpha.$$

Now suppose $\Delta(f) \neq 0$. Since $\text{Res}(f, h) = -\Delta(f)^2 \neq 0$ and $\Delta(h) = -3\Delta(f) \neq 0$, we have $f(x) = (x - \alpha_1)(x - \alpha_2)(x - \alpha_3)$, $h(x) = 3a(x - \beta_1)(x - \beta_2)$, for some $\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2$ which are pairwise distinct. Write

$$T_f(x) = x - \frac{f(x)h(x)}{h(x)f'(x) - f(x)h'(x)}. \tag{2}$$

Observe $h(\alpha_i)f'(\alpha_i) \neq 0$ and $f(\beta_j)h'(\beta_j) \neq 0$. This tells us the terms $f(x)h(x)$ and $h(x)f'(x) - f(x)h'(x)$ have no common factors. By expanding (1), it is therefore clear that T_f is a quartic rational map. □

Corollary 2. *Let f be as above, and $\Delta(f) \neq 0$. If $a \neq 0$, then the fixed points of T_f are the roots of f and the roots of h. If $a = 0$, then the fixed points of T_f are the roots of f, the root of h and ∞.*

Proof. By Proposition 1, as a rational map of degree 4, T_f has at most 5 fixed points. By (2), any roots of f and h are fixed points. These roots are pairwise distinct, since all of $\Delta(f)$, $\Delta(h)$ and $\text{Res}(f, h)$ are nonzero.

If $a \neq 0$, then $\deg f = 3$ and $\deg h = 2$, so we have found all fixed points of T_f. If $a = 0$, then h is linear, so we have only found 4 fixed points. However, it is clear from (1) that ∞ is a fixed point of T_f in this case. □

From now on, assume the following to be fixed.

K	Number field with ring of integers $\mathcal{O}_K$.
f	Cubic polynomial with coefficients in K. Assume $\Delta(f) \neq 0$ and write $\alpha_1, \alpha_2, \alpha_3$ for the roots of f. We also assume $\alpha_3 = -(\alpha_1 + \alpha_2)$.
h	Quadratic polynomial defined as in Section 1. Write β_1, β_2 for its roots.
T_f	McMullen map for f defined as in (1).
x_0	Element of K.
(x_n)	Sequence defined by $x_{n+1} = T_f(x_n)$; assume it is not eventually periodic.

Proposition 3. *Let S_∞ be the finite set of prime ideals $\mathfrak{p}$ of $\mathcal{O}_K$ such that $\mathrm{ord}_\mathfrak{p}(\alpha) < 0$ for some root α of f; or*

- *if $a \neq 0$, $\mathfrak{p} \mid 3a$; or $\mathrm{ord}_\mathfrak{p}(\beta) < 0$ for some root β of h.*
- *if $a = 0$, $\mathfrak{p} \mid 6b$.*

The sequence (x_n) does not converge to ∞ in $\mathbb{P}^1(K_\mathfrak{p})$ for any $\mathfrak{p}$ outside S_∞.

Proof. First suppose $a \neq 0$. Define $D(x) = (f'(x)h(x) - h'(x)f(x))/3a$. Then we can rewrite

$$T_f(x) = x - \frac{f(x)h(x)}{3aD(x)} = x - \frac{(x-\alpha_1)(x-\alpha_2)(x-\alpha_3)(x-\beta_1)(x-\beta_2)}{D(x)}.$$

Define its reciprocal polynomial to be

$$D^*(x) = x^4 D\left(1/x\right)$$

and observe $D^*(0) = 1$. Hence,

$$T_f(1/x) = \frac{D^*(x) - (1-\alpha_1 x)\cdots(1-\beta_2 x)}{xD^*(x)}.$$

Fix $\mathfrak{p} \notin S_\infty$ and suppose x_n is such that $\mathrm{ord}_\mathfrak{p}(x_n) = \ell < 0$. Let $y_n = 1/x_n$. We have

$$x_{n+1} = T_f(1/y_n) = \frac{D^*(y_n) - (1-\alpha_1 y_n)\cdots(1-\beta_2 y_n)}{y_n D^*(y_n)},$$

hence

$$x_{n+1}y_n = \frac{D^*(y_n) - (1-\alpha_1 y_n)\cdots(1-\beta_2 y_n)}{D^*(y_n)} \equiv \frac{1-1}{1} = 0 \mod \mathfrak{p}.$$

This implies $\mathrm{ord}_\mathfrak{p}(x_{n+1}) \geq \ell + 1$. In particular, this means (x_n) cannot converge to ∞ in $K_\mathfrak{p}$.

Suppose now $a = 0$. Note $h(x) = 9bx$ and $f(x) = x^3 + b$. Define $D(x) = [f'(x)h(x) - h'(x)f(x)]/9b = 2x^3 - b$. Its reciprocal polynomial is $D^*(x) = x^3 D(1/x) = 2 - bx^3$, and $D^*(0) = 2$. Mimicking the above passages, fixing $\mathfrak{p} \notin S_\infty$ we obtain

$$x_{n+1}y_n = \frac{D^*(y_n) - (1-\alpha_1 y_n)(1-\alpha_2 y_n)(1-\alpha_3 y_n)}{D^*(y_n)} \equiv \frac{2-1}{2} \equiv \frac{1}{2} \mod \mathfrak{p}$$

Therefore we can conclude $\mathrm{ord}_\mathfrak{p}(x_{n+1}) = \mathrm{ord}_\mathfrak{p}(x_n) = \ell$. By induction, this means (x_n) cannot converge to ∞ in $K_\mathfrak{p}$. □

Remark 4. *In fact, we have shown something stronger than this. For $a \neq 0$, we have seen that, if $\mathrm{ord}_\mathfrak{p}(x_n) = -1$ then x_{n+1} is integral with respect to $\mathfrak{p}$. This means that, if T_f has good reduction at $\mathfrak{p}$, if $\mathrm{ord}_\mathfrak{p}(x_n) = l$ for any $l < 0$, then x_{n+1} is integral with respect to $\mathfrak{p}$. In particular, we know $x_{n+1} \equiv 3b/a \pmod{\mathfrak{p}}$.*

Corollary 5. *Suppose $\mathfrak{p}$ is a prime ideal of $\mathcal{O}_K$ such that $\mathfrak{p} \notin S_\infty$. If (x_n) converges to $\gamma \in \mathbb{P}^1(K_\mathfrak{p})$, then γ is a root of f or a root of h.*

Proof. By Proposition 3 we see that $\gamma \neq \infty$. It is easy to check, in either of the cases $a \neq 0$ and $a = 0$, that $D(\zeta) \neq 0$ for any root ζ of f or h. For $a \neq 0$, we have

$$x_{n+1} = x_n - \frac{(x_n - \alpha_1)\cdots(x_n - \beta_2)}{D(x_n)}.$$

Letting $n \to \infty$ and subtracting γ from both sides yields

$$\frac{(\gamma - \alpha_1)\cdots(\gamma - \beta_2)}{D(\gamma)} = 0,$$

from which the result follows. The case $a = 0$ follows similarly. $\square$

For the following theorems, we use the following result of Ingram and Silverman. For the statement, recall that if (y_n) is a sequence of nonzero elements of K, we say $\mathfrak{p}$ is a primitive prime factor of the numerator of y_n if $\operatorname{ord}_{\mathfrak{p}}(y_n) > 0$ but $\operatorname{ord}_{\mathfrak{p}}(y_m) = 0$ for all $m < n$.

Theorem 6. *Let $\phi \in K(t)$ be a rational function of degree at least 2, let $\gamma \in K$ be a periodic point for ϕ, and let $y_0 \in K$ be a point with infinite ϕ-orbit; i.e., the sequence defined by $y_{n+1} = \phi(y_n)$ for $n \geqslant 0$ is not eventually periodic. Then for all sufficiently large n, the element $y_n - \gamma$ has a primitive prime factor in its numerator if and only if ϕ is not totally ramified at γ.*

Theorem 7. *For a root α of f, the sequence (x_n) converges v-adically to α for infinitely many places v of K.*

Proof. We can rewrite our expression for T_f in the following way.

$$\begin{aligned}
T_f(x) &= x - \frac{q(x)}{q'(x)} \\
&= x - \frac{(x-\alpha)g_\alpha(x)h(x)}{h(x)[g_\alpha(x) + (x-\alpha)g'_\alpha(x)] - (x-\alpha)g_\alpha(x)h'(x)} \\
&= \alpha + (x-\alpha)\left(1 - \frac{g_\alpha(x)h(x)}{h(x)[g_\alpha(x) + (x-\alpha)g'_\alpha(x)] - (x-\alpha)g_\alpha(x)h'(x)}\right) \\
&= \alpha + (x-\alpha)^2\left(\frac{h(x)g'_\alpha(x) - g_\alpha(x)h'(x)}{h(x)g_\alpha(x) + (x-\alpha)[h(x)g'_\alpha(x) - g_\alpha(x)h'(x)]}\right) \\
&= \alpha + (x-\alpha)^3\left(\frac{k_\alpha(x)}{h(x)g_\alpha(x) + (x-\alpha)^2 k_\alpha(x)}\right). \quad (*)
\end{aligned}$$

Define S_α to be the following set of prime ideals of $\mathcal{O}_K$,

$$S_\alpha = \left\{ \mathfrak{p} \;\middle|\; \operatorname{ord}_{\mathfrak{p}}\left(\frac{h(\alpha)g_\alpha(\alpha)}{k_\alpha(\alpha)}\right) \geqslant 2 \right\}.$$

Given $\mathfrak{p} \notin S_\alpha$, if $\mathfrak{p}^m \mid x_n - \alpha$ for some $n \geqslant 0$ and some $m > 0$, then $(*)$ implies that $\mathfrak{p}^{3m-1} \mid x_{n+1} - \alpha$, so we have $\mathfrak{p}$-adic convergence of (x_n) to α. By Theorem 6, there are infinitely many such $\mathfrak{p}$ if and only if T_f is not totally ramified at α.

Recall $k_\alpha(x) = 3(3b - a\alpha)x - (9a\alpha^2 - 9b\alpha + 8a^2)$. By $(*)$, since T_f is a quartic rational map, we have that T_f is totally ramified at α if and only if $k_\alpha(x) = C(x - \alpha)$ for some constant C. Indeed, we have

$$\begin{aligned} k_\alpha(x) = C(x-\alpha) &\iff \frac{9a\alpha^2 - 9b\alpha + 8a^2}{3(3b - a\alpha)} = \alpha, \\ &\iff 6a\alpha^2 - 9b\alpha + 4a^2 = 0. \end{aligned}$$

Knowing α is a root of f, we compute

$$\operatorname{Res}(x^3 + ax + b, 6ax^2 - 9bx + 4a^2) = \Delta(f)^2 \neq 0,$$

so T_f cannot be totally ramified at α. This completes the proof. $\square$

Theorem 8. *For a root β of h, the sequence (x_n) does not converge $\mathfrak{p}$-adically to β for any $\mathfrak{p}$, except possibly if $\mathfrak{p}$ lies over 2.*

Proof. We suppose $a \neq 0$. (The case $a = 0$ is very similar, but slightly less algebraically intensive.) WLOG $\beta = \beta_1$. Then,

$$\begin{aligned} T_f(x) &= x - \frac{f(x)h(x)}{h(x)f'(x) - f(x)h'(x)} \\ &= x - \frac{3a(x-\beta_1)(x-\beta_2)f(x)}{3a(x-\beta_1)(x-\beta_2)f'(x) - 3af(x)[(x-\beta_1)+(x-\beta_2)]} \\ &= x - \frac{(x-\beta_1)(x-\beta_2)f(x)}{(x-\beta_1)[(x-\beta_2)f'(x) - f(x)] - (x-\beta_2)f(x)} \\ &= \beta_1 + (x-\beta_1)\left(1 - \frac{(x-\beta_2)f(x)}{(x-\beta_1)[(x-\beta_2)f'(x) - f(x)] - (x-\beta_2)f(x)}\right) \\ &= \beta_1 + (x-\beta_1)\left(\frac{(x-\beta_1)[(x-\beta_2)f'(x) - f(x)] - 2(x-\beta_2)f(x)}{(x-\beta_1)[(x-\beta_2)f'(x) - f(x)] - (x-\beta_2)f(x)}\right) \end{aligned}$$

Suppose $\mathfrak{p}^\ell \| (x - \beta_1)$. It is clear that if $\mathfrak{p} \nmid 2$, then $\mathfrak{p}^\ell \| (T_f(x) - \beta_1)$. In particular, (x_n) cannot converge to β_1 in $\mathbb{P}^1(K_\mathfrak{p})$. $\square$

Theorem 9. *The sequence (x_n) diverges in $\mathbb{P}^1(K_v)$ for infinitely many places v of K.*

Proof. Choose γ an unramified periodic point of T_f with period $q > 1$. Suppose $\mathfrak{p}$ is a prime factor of $x_n - \gamma$ for some n, and suppose further that T_f has good reduction at $\mathfrak{p}$, that $\mathfrak{p}$ does not divide the numerator or denominator of $\gamma - \alpha$ for any root α of f, that $\mathfrak{p} \notin S_\infty$ and that $\mathfrak{p}$ does not lie over 2. Then

$$x_{n+q} = \underbrace{N \circ \cdots \circ N}_{q \text{ times}}(x_n) \equiv \underbrace{N \circ \cdots \circ N}_{q \text{ times}}(\gamma) = \gamma \pmod{\mathfrak{p}}.$$

By induction, we find that $x_{n+kq} \equiv \gamma \pmod{\mathfrak{p}}$ for each $k \geqslant 0$. In particular, this shows that $x_{n+kq} \not\equiv \alpha \pmod{\mathfrak{p}}$ for any $k \geqslant 0$, and hence (x_n) does not converge to α in the $\mathfrak{p}$-adic topology. Since $\mathfrak{p} \notin S_\infty$ and $\mathfrak{p}$ does not lie over 2, we also know by Proposition 3 and Theorem

8 that (x_n) does not converge to ∞ or to a root of h in the $\mathfrak{p}$-adic topology. Therefore, by Corollary 5, (x_n) diverges in the $\mathfrak{p}$-adic topology.

By Theorem 6, we see that $x_n - \gamma$ has a primitive prime factor for each sufficiently large n, and so the above argument succeeds for infinitely many prime ideals $\mathfrak{p}$. $\square$

Lastly, it is straightforward to show that the set of primes yielding $\mathfrak{p}$-adic divergence has positive lower density, once we quote the following special case of Thereom 5 of Bendetto et al (2012).

Theorem 10. *Let K be a number field and $\varphi \in K(x)$ be a rational function of degree at least 2. Let $x_0 \in \mathbb{P}^1(K)$ be a point with infinite φ-orbit, and let $\mathcal{B} \subset \mathbb{P}^1(K)$ be a finite set of preperiodic points for φ. Then the set of non-archimedean places $\mathcal{P}$ such that $\varphi^n(x_0) \not\equiv \beta \pmod{\mathfrak{p}}$ for any $n \geqslant 0$, any $\beta \in \mathcal{B}$, and any $\mathfrak{p} \in \mathcal{P}$, has positive lower density (when ordered by norm).*

Theorem 11. *The set of places v of K for which (x_n) diverges v-adically has positive lower density.*

Proof. In Thereom 10, set $\varphi = T_f$ and let $\mathcal{B}$ be the set of roots of f. Thereom 10 implies that the sequence (x_n) fails to $\mathfrak{p}$-adically approach $\mathcal{B}$ for a set of primes $\mathfrak{p}$ of K with positive lower density. We have proved in Proposition 3 and Theorem 8 that (x_n) converges $\mathfrak{p}$-adically to a root of h or to ∞ for finitely many places v. Applying Corollary 5 implies that (x_n) diverges in the $\mathfrak{p}$-adic topology for a set of primes $\mathfrak{p}$ with positive lower density. $\square$

3 Examples

Example 12. *Let us now consider $a = 0$ and $b = -1$, so*

$$f(x) = x^3 - 1 \quad \textit{and} \quad T_f(x) = \frac{x(2 + x^3)}{1 + 2x^3}.$$

Since $a = 0$, reducing the map modulo p and checking if $f(x_n) \equiv 0 \mod p$ is necessary and sufficient for convergence. Hence (a variation of) our previous computational method still works and we can conduct similar analyses.

Note we need to exclude 3 because of bad reduction and 2 because $S_\infty = \{2\}$, while $S_\alpha = \varnothing$. If we proceed with a linear regression on the data $(\log(X), \log(\delta(X))$ for X ranging up to 2,000,000 in increments of 20,000, we obtain $m = -0.5032$, $R^2 = 0.9950$.

Example 13. *Let $b = 0$, so 0 is a root of f. Then T_f takes the form*

$$T_f(x) = \frac{-8a^2x^3}{3ax^4 - 6a^2x^2 - a^3}.$$

Suppose $\mathrm{ord}_{\mathfrak{p}}(x_n) = l < 0$. Unless $\mathrm{ord}_{\mathfrak{p}}(3/a) > 0$, this expression tells us that $\mathrm{ord}_{\mathfrak{p}}(x_{n+1}) \geqslant -l > 0$. Therefore, by the proof of Theorem 7, (x_n) converges to 0 in the $\mathfrak{p}$-adic topology.

Altogether, this means we have a complete test for convergence in the case $b = 0$: reduce the map modulo p, then we have p-adic convergence if and only if the map eventually reaches either ∞ or a root of $(f \mod p)$.

For example, let $a = 3$ and $b = 0$, so

$$f(x) = x^3 + 3x \quad \text{and} \quad T_f(x) = -\frac{8x^3}{x^4 - 6x^2 - 3}.$$

We choose $x_0 = 2$ as our starting point. We exclude 2 and 3 because of bad reduction, while $S_\infty = S_\alpha = \varnothing$. If we proceed with a linear regression on the data $(\log(X), \log(\delta(X))$ for X ranging up to 2,000,000 in increments of 20,000, we obtain $m = -0.5067$ and $R^2 = 0.9978$.

Example 14. *When a and b are both nonzero, we must work slightly harder to get a complete test for p-adic convergence. For example, let $a = -4$ and $b = 3$, so 1 is a root of f. Then*

$$f(x) = x^3 - 4x + 3 \quad \text{and} \quad T_f(x) = \frac{-27x^4 + 128x^3 - 216x^2 + 162x - 48}{12x^4 - 54x^3 + 96x^2 - 72x + 17}.$$

Since a and b are nonzero, if $\mathrm{ord}_{\mathfrak{p}}(x_n) < 0$, the $\mathfrak{p}$-adic convergence of (x_n) depends on the choice of $\mathfrak{p}$. For $\mathfrak{p} \nmid 2$, we know that x_{n+1} is integral; in fact, we have $x_{n+1} \equiv -9/4 \pmod{\mathfrak{p}}$, so we can restart our original convergence test from this point. We illustrate this dependence with two different choices of starting point and prime.

Choose $x_0 = 3$ as our starting point, and let $p = 179$. One can compute that $\mathrm{ord}_{179}(x_1) = -1$. It turns out that $\mathrm{ord}_{179}(x_{10} - 1) \geqslant 1$, so by the proof of Theorem 7, we in fact have 179-adic convergence of (x_n) to the root 1 of f.

On the contrary, if we choose $x_0 = 4$ as our starting point and let $p = 881$, then also $\mathrm{ord}_{881}(x_1) = -1$, but this time (x_n) diverges in the 881-adic topology, because the reduction $(x_n \mod p)$ for $n \geqslant 2$ eventually reaches a cycle.

This gives us a complete p-adic convergence test. We would like to create similar data to in Examples 12 and 13. We choose $x_0 = 2$ as our starting point. We exclude 3 and 13 because they give bad reduction of T_f, while $S_\infty = \{2, 3\}$ and $S_\alpha = \{13\}$. We proceed with the same log-log *linear regression as before and obtain $m = -0.5161$ and $R^2 = 0.9987$.*

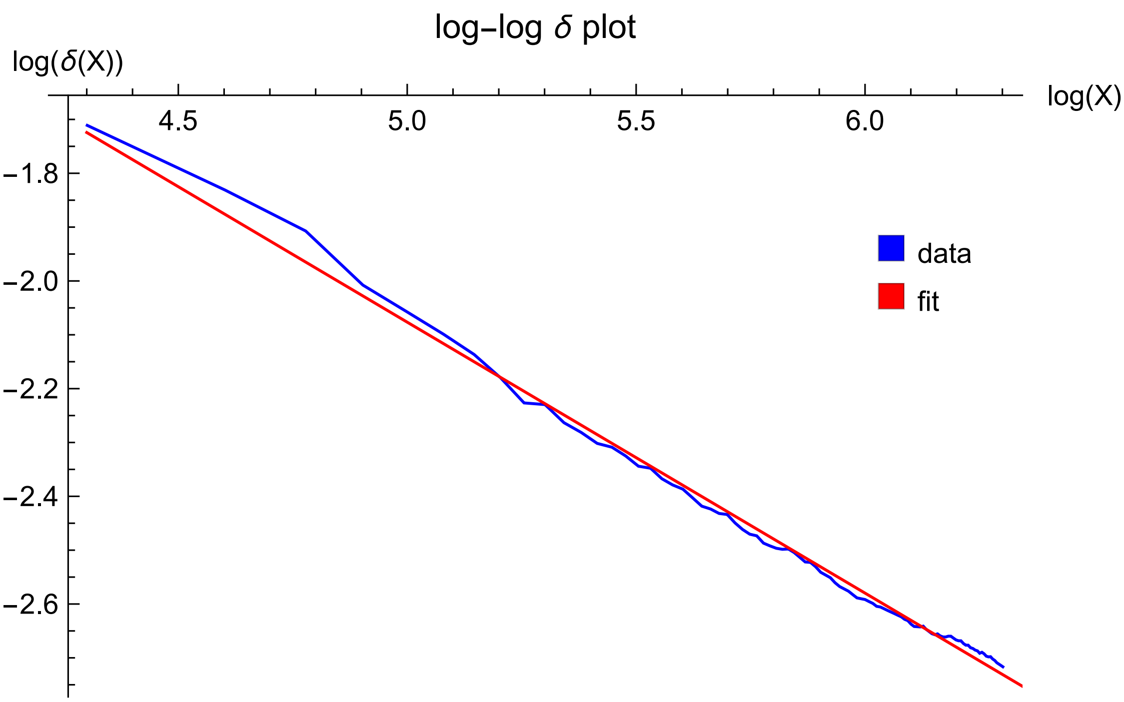

Figure 1: Linear Regression for Example 12.

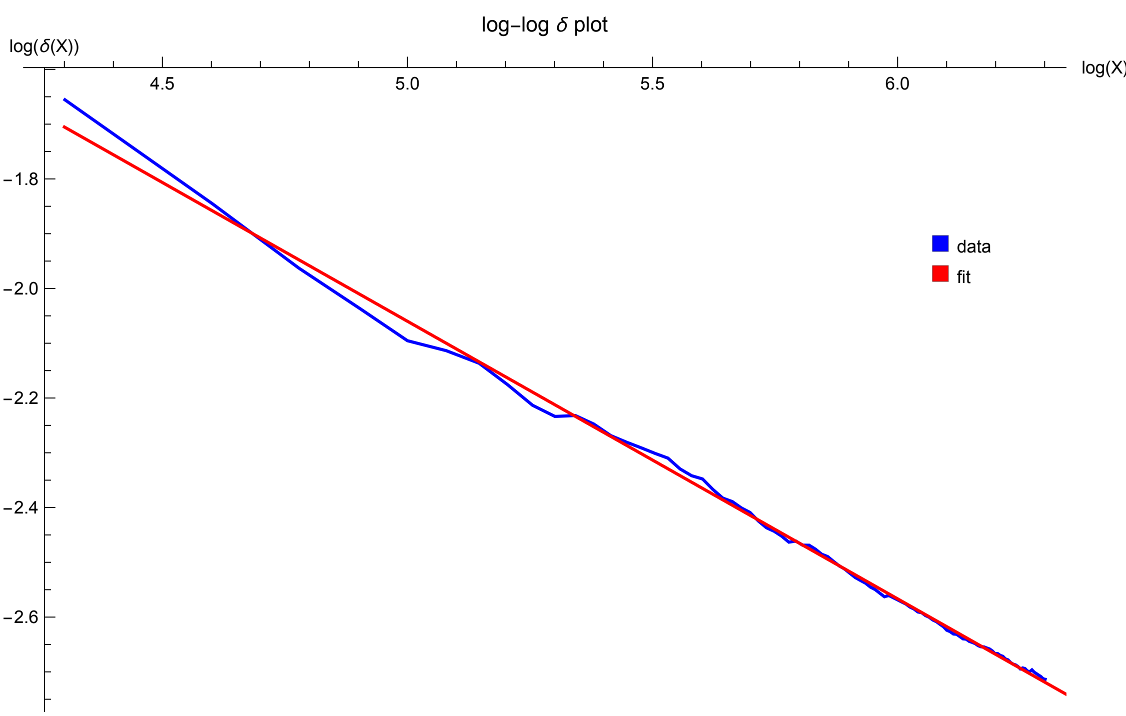

Figure 2: Linear Regression for Example 13.

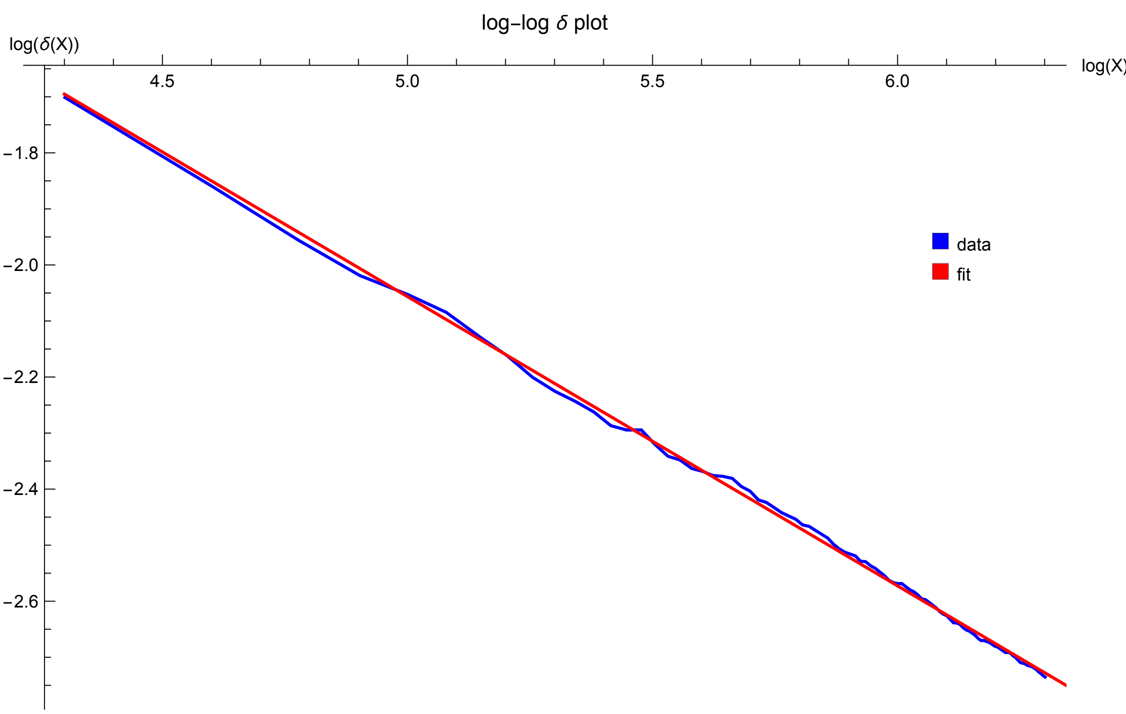

Figure 3: Linear Regression for Example 14.